4 A simple square demonstration

Section 3 established how to compare solutions of the supported system. We now apply that protocol to one square, first with a mesh small enough to inspect its equations, then with a larger mesh on which iterative methods behave differently. Both use the same material, loading and supports. This is a controlled demonstration, not a selection of the objects for the later study.

4.1 The square and its small system

The square occupies $[0,1] \times [0,1]$ metres, with thickness $t = 0.01$ m, Young's modulus $E = 200$ GPa and Poisson's ratio $\nu = 0.3$. We use Section 2's plane-stress model. The whole left edge is fixed in both directions. A uniform rightward traction $p = 100$ MPa acts on the right edge; the top and bottom are traction-free and there are no body forces. Thus the line load is $tp = 10^6$ N/m and the total applied force is 10^6 N, not a prescribed displacement. These are the repository's force-driven example settings [1].

The small mesh has two cells in each direction, each split from its bottom-left to its top-right corner, giving nine nodes and eight triangles (Fig. 1a). Nodes 0, 3 and 6 are fixed. Using Section 2's interleaved component order, the remaining unknowns are

$$x = (u_{x,1}, u_{y,1}, u_{x,2}, u_{y,2}, u_{x,4}, u_{y,4}, u_{x,5}, u_{y,5}, u_{x,7}, u_{y,7}, u_{x,8}, u_{y,8})^T. \tag{1}$$

Each loaded half-metre edge contributes equally to its endpoints, so nodes 2, 5 and 8 receive horizontal forces of 250, 500 and 250 kN. Consequently, the reduced system has a 12×12 matrix and

$$b = 10^6 \,\mathrm{N}\,(0, 0, 0.25, 0, 0, 0, 0.50, 0, 0, 0, 0.25, 0)^T. \tag{2}$$

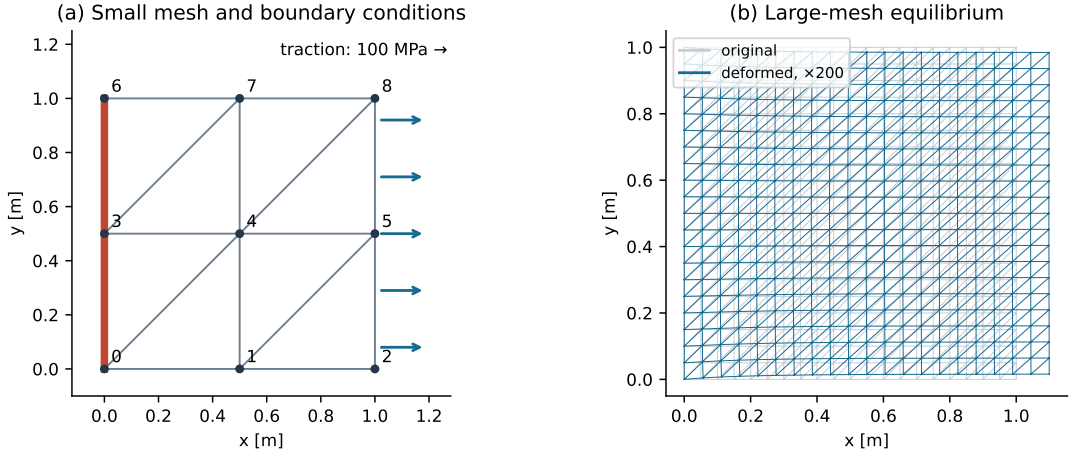


Figure 1: The same clamped, traction-loaded square on two meshes. (a) Node numbers identify the small system's unknowns. (b) The 800-triangle direct-reference solution is overlaid on the original mesh. Coordinates are in metres; displacement is amplified by 200 for visibility, not for the calculation.

The matrix couples horizontal and vertical movements rather than representing independent springs. For example, dividing its first row by $10^9/91$ N/m gives the force-balance equation at node 1,

$$\begin{aligned} 270u_{x,1} - 65u_{y,1} - 100u_{x,2} + 30u_{y,2} \\ - 70u_{x,4} + 65u_{y,4} - 65u_{y,5} = 0. \end{aligned} \tag{3}$$

We solve all twelve equations together using the Cholesky steps from Section 3. Table 1 gives the resulting free components. The complete A, b, Cholesky factor, intermediate vector and solution are supplied as CSV files with the source, so the worked system can be reproduced without rounding the displayed values.

Table 1: Small-mesh Cholesky solution, rounded to five decimal places. All components at the three fixed nodes are zero.

Node	1	2	4	5	7	8
u_x [mm]	0.26135	0.51119	0.23163	0.48215	0.22445	0.46780
u_y [mm]	0.07606	0.10453	0.00749	0.02937	-0.05413	-0.04609

The supported matrix is positive definite, with $\kappa_2(A) \approx 95.8$. Cholesky, pivoted LU and explicit inversion all give relative residuals below 8×10^{-16}; their differences from the Cholesky solution are below 5×10^{-16} in relative norm. Inversion is included only for this tiny example, not as the procedure for larger meshes.

4.2 Iterative solutions on a larger mesh

Keeping the physical case fixed, we use 20 cells per direction: 441 nodes, 800 triangles and 840 free components. Table 2 reports a fresh execution of the Section 3 protocol, including its zero initial guess, 20 000-iteration budget and strict true-residual target $\rho < 10^{-8}$. Sparse LU provides the untimed same-mesh reference. Times are the minimum of three setup-inclusive calls, with histories collected separately; machine details and all repeats accompany the data.

Table 2: Large-square results. ρ is the relative residual and e_{ref} the relative difference from sparse LU, as defined in Section 3. "Cap" and "guard" indicate unsuccessful runs, so their times are not times to the requested accuracy. Acceptance uses unrounded residuals.

Method	Iterations	ρ	e_{ref}	Time [s]	Outcome
Jacobi	49	1.324×10^{10}	3.378×10^{7}	0.000559	Guard
Damped Jacobi	20000	4.758×10^{-6}	8.727×10^{-5}	0.227	Cap
Gauss–Seidel	19136	9.996×10^{-9}	1.740×10^{-7}	5.94	Pass
SD	20000	3.081×10^{-6}	4.221×10^{-5}	0.384	Cap
CG	242	9.507×10^{-9}	9.793×10^{-10}	0.00431	Pass
GMRES (full)	228	9.476×10^{-9}	5.965×10^{-9}	0.146	Pass
MINRES	236	9.495×10^{-9}	5.985×10^{-9}	0.01	Pass
Diagonal PCG	226	9.014×10^{-9}	5.014×10^{-10}	0.00499	Pass

CG and diagonal PCG reach the target in 242 and 226 steps respectively, whereas Gauss–Seidel needs 19 136 sweeps. Ordinary Jacobi diverges; damping prevents that rapid growth here but does not reach the target within the budget. Figure 2 connects these residual histories to selected CG displacement approximations. The iterates approach one static solution, not successive physical shapes in time. Fewer steps also need not mean a faster implementation: orthogonalization and setup costs differ between methods. This single case does not establish a general solver ranking.

4.3 Equilibrium checks and limits

In the large-mesh direct solution, right-edge horizontal displacements range from approximately 0.492 to 0.497 mm. The clamp prevents movement at the left edge while the material contracts sideways (Fig. 1b). The full displacement has exactly zero prescribed components. Using only

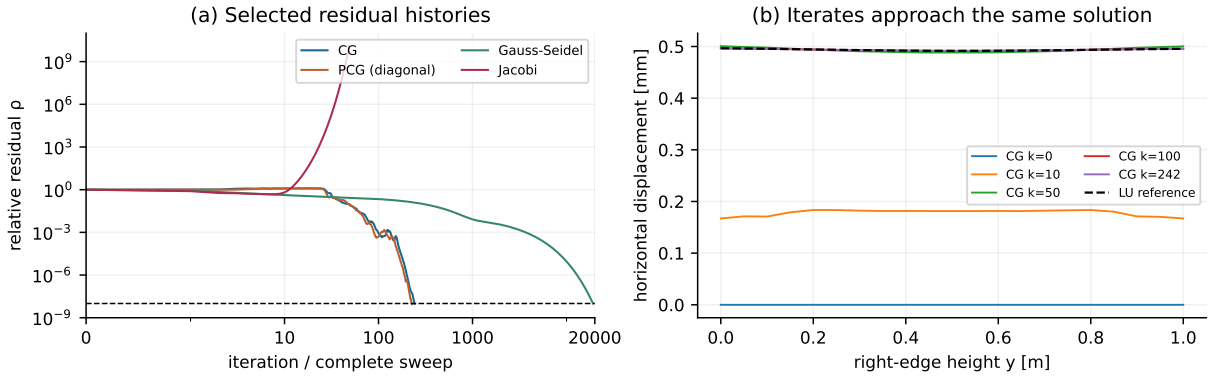


Figure 2: Large-square convergence. (a) Selected true relative residuals, including the failed Jacobi run; the iteration axis is logarithmic beyond the first step. The dashed line is the acceptance threshold. (b) Horizontal displacement along the right edge at selected CG iterations k, without amplification. The final curve overlaps the LU reference; residuals need not decrease at every CG step.

constrained components for reactions, the summed support force is approximately $(-10^6, 0)$ N; the remaining net-force norm is below 4×10^{-9} N and the moment imbalance about the lower-left corner is below 2×10^{-9} N m. The reference's relative residual is 3.83×10^{-14}.

These checks support the assembly and algebraic solution of this case. They do not establish continuum accuracy: the coarse mesh's unequal corner movements also warn against interpreting its deformation as a mesh-independent prediction. A systematic refinement study and comparisons between different objects remain for the later sections. The team's assembly implementation is used here; permission to substitute it for the course-provided routine remains unresolved, as noted in Section 2.

References

[1] NLA project team, *NLA: force-driven square and solver benchmark*, `https://github.com/Pueblo98/NLA`, 2026, Source snapshot 9736c8e; unchanged implementation files accompany the section's reproducibility package.